

Shihao(Rex) Ma

CONTACT INFORMATION	108 College St Toronto, ON M5G 0C6	rex.ma@mail.utoronto.ca google scholar , personal website
EDUCATION	University of Toronto , Ontario, Canada Ph.D. in Computer Sciences - Supervisory Committee: Bo Wang, Anna Goldenberg, Benjamin Haibe-Kains - Research Focus: Multimodal integration, genomic foundational model, LLM in biology	Sept. 2020 - Aug. 2026
	University of Toronto , Ontario, Canada B.A.Sc in Computer Engineering, GPA: 3.87/4.00	Sept. 2015 - June 2020
EXPERIENCE	Xaira Therapeutics , South San Francisco, United State <i>AI Scientist Intern</i> Agentic AI for protein design and target identification, scRNA-seq, perturb-seq foundation modeling	Sept. 2025 - Sept. 2026
	Vector Institute for Artificial Intelligence , Toronto, Canada <i>Machine Learning Researcher</i>	Sept. 2020 - Aug. 2026
	Fable Therapeutics , Toronto, Canada <i>Research Scientist Intern</i> De novo antibody protein design, 3D geometric deep learning, protein structure generation.	May 2022 - Aug. 2022
	University Health Network , Toronto, Canada <i>Machine Learning Engineering Intern</i> Prognosis prediction of patients with heart failure, deep learning for single-cell RNA-seq data.	May 2019 - Aug. 2019
	IBM - DB2 Availability & Recovery Domain , Toronto, Canada <i>Software Engineering Intern</i> - Created a prototype for Goldman Sachs to improve exclusive database access for Db2 offline utility commands. - Addressed over 60 recovery defects in the DB2 code base.	May 2017 - May 2018
PUBLICATIONS	Wang, C.*, Karimzadeh, M.*, Ravindra, N.*, Bounds, L.*, Alerasool, N.*, Huang, A.*, Ma, S. et al. X-cell: Scaling causal perturbation prediction across diverse cellular contexts via diffusion language models <i>bioRxiv</i> (2026). [Paper] Ma, S. et al. Moving towards genome-wide data integration for patient stratification with Integrate Any Omics. <i>Nature Machine Intelligence</i> , 7, p29-42 (2025). [Paper] Xu, Y.*, Ma, S.*, Cui, H*. et al. AGILE platform: a deep learning powered approach to accelerate LNP development for mRNA delivery. <i>Nature Communication</i> , 15, 6305 (2024). [Paper] Chen, X.*, Ma, S.* et al. Ctrl-DNA: Controllable Cell-Type-Specific Regulatory DNA Design via Constrained RL. <i>NeurIPS Spotlight Presentation (top 3.2%)</i> (2025). [Paper] Douglas L.*, Ma, S.* et al. Predictors of mortality among long-term care residents with SARS-CoV-2 infection. <i>Journal of the American Geriatrics Society</i> , 69, p3377-3388 (2021). [Paper]	

Fallahpour, A.*, Magnuson, A.*, Gupta, P.*, Ma, S. et al. BioReason: Incentivizing Multimodal Biological Reasoning within a DNA-LLM Model. *NeurIPS Poster Presentation* (2025). [[Paper](#)]

Abdul, K., Ma, S. et al. Comparison of machine learning and conventional statistical modeling for predicting readmission following acute heart failure hospitalization *American Heart Journal*, 277, p93–103 (2024). [[Paper](#)]

Austin, D., Douglas L., Wang C., Ma, S. et al. Comparison of machine learning and the regression-based EHMGR model for predicting early mortality in acute heart failure *International Journal of Cardiology*, 65, p78–84 (2022). [[Paper](#)]

Douglas L., Wang C., McAlister F. Ma, S. et al. Factors associated with SARS-CoV-2 test positivity in long-term care homes: a population-based cohort analysis using machine learning *The Lancet Regional Health–Americas*, 6, 100146 (2022). [[Paper](#)]

Ma, J., Zhang Y., Gu S., Ge Cheng., Ma, S. et al. Unleashing the strengths of unlabelled data in deep learning-assisted pan-cancer abdominal organ quantification: the FLARE22 challenge *The Lancet Digital Health*, Volume 6, Issue 11, e815 - e826 (2024). [[Paper](#)]

Nitski, O., Azhie A., Qazi F., Wang X., Ma, S. et al. Long-term mortality risk stratification of liver transplant recipients: real-time application of deep learning algorithms on longitudinal data. *The Lancet Digital Health*, Volume 3, Issue 5, e295 - e305 (2021). [[Paper](#)]

PATENTS Systems and Methods for AI-Based Ionizable Lipid Development for Lipid Nanoparticle (LNP)-Based Cargo Molecule Delivery (PCT/CA2024/050644).

HONORS AND AWARDS	University of Toronto Computer Science Departmental Fellowship	2023-2024
	University of Toronto Computer Science Departmental Fellowship	2022-2023
	Undergraduate Research Opportunity Program (UROP) Awards	2018

TEACHING EXPERIENCE	University of Toronto , Toronto, Canada	
	<i>Teaching Assistant</i>	
	CSC413/CSC2516 - Neural Networks and Deep Learning	Fall 2024
	CSC108 - Introduction to Computer Programming	Winter 2024
	CSC413/CSC2516 - Neural Networks and Deep Learning (Head TA)	Winter 2023
	CSC413/CSC2516 - Neural Networks and Deep Learning	Winter 2022
	CSC110 - Foundations of Computer Science	Fall 2021
	CSC413/CSC2516 - Neural Networks and Deep Learning	Winter 2021
	CSC108 - Introduction to Computer Programming	Fall 2020

COMPUTER SKILLS	• Proficient with: Python, C++, R, Git, Linus, Agile (Scrum) • Familiar with: HTML5, SQL, CSS, Java, Matlab • Soft skills: Public Speaker, Motivator, Collaborator, Leader
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